

Step 1: regularizing coefficients of A: We follow [AN24]. Let $\theta \in \mathcal{D}(\mathbb{R})$ a nonnegative function with $\int_{\mathbb{R}} \theta(t) dt = 1$. For all $p \geq 1$, let $\theta_p(t) = p\theta(pt)$ be the associated mollifying sequence. We set $A_p(t, x) := (\theta_p \star A(\cdot, x))(t)$, i.e., we mollify the matrix-valued function A in the time variable only. For all $p \geq 1$ and $t \in \mathbb{R}$, we set

$$B_t^p(u, v) := \int_{\mathbb{R}^n} \omega^{-1} A_p(t, \cdot) \nabla_x u \cdot \overline{\nabla_x v} \, d\omega + \frac{1}{p} \langle u, v \rangle_{2, \omega}.$$

We check easily that $\min(1/l, \nu) \|u\|_{H_{\omega}^1(\mathbb{R}^n)}^2 \leq \operatorname{Re}(B_t^p(u, u))$ and $\operatorname{Im}(B_t^p(u, u)) \leq \frac{M}{\nu} \operatorname{Re}(B_t^p(u, u))$. In particular, the quadratic form of $\operatorname{Re}(B_t^p(\cdot, \cdot))$ is closed. Moreover, we have

$$|B_t^p(u, u) - B_s^p(u, u)| \leq M \left\| \frac{d\theta_p}{dt} \right\|_{L^1(\mathbb{R})} |t - s| \|\nabla_x u\|_{2, \omega}^2 \leq \frac{pM \|\dot{\theta}\|_{L^1(\mathbb{R})}}{\nu} |t - s| \operatorname{Re}(B_t^p(u, u)),$$

where $\dot{\theta}$ is the derivative of θ . For all $p \geq 1$, we set $U_p(t) := \Gamma_p(t, s)f$ where Γ_p is the fundamental solution of the parabolic operator associated to the family $(B_t^p)_{t \in \mathbb{R}}$. Combining [Kat61, Theorem III] with uniqueness in $L^2((s, \mathfrak{T}); H_{\omega}^1(\mathbb{R}^n))$ for any $\mathfrak{T} > s$, we have for all $p \geq 1$, $U_p : (s, \infty) \rightarrow L_{\omega}^2(\mathbb{R}^n)$ is strongly differentiable. Note that U_p is a real-valued function by the same argument as we did for U . Since $\nabla_x U_p(t) \in L_{\omega}^2(\mathbb{R}^n)$, we have $\partial_t |U_p(t)|, \nabla_x |U_p(t)| \in L_{\omega}^2(\mathbb{R}^n)$ with

$$\partial_t |U_p(t)| = \begin{cases} \partial_t U_p(t) & \text{if } U_p(t) \geq 0, \\ -\partial_t U_p(t) & \text{if } U_p(t) < 0, \end{cases} \quad \text{and} \quad \nabla_x |U_p(t)| = \begin{cases} \nabla_x U_p(t) & \text{if } U_p(t) \geq 0, \\ -\nabla_x U_p(t) & \text{if } U_p(t) < 0. \end{cases}$$

Using this, we have

$$\begin{aligned} -\frac{d}{dt} \langle U_p(t) - |U_p(t)|, U_p(t) - |U_p(t)| \rangle_{2, \omega} &= -2 \langle \partial_t (U_p(t) - |U_p(t)|), U_p(t) - |U_p(t)| \rangle_{2, \omega} \\ &= -4 \langle \partial_t U_p(t), U_p(t) - |U_p(t)| \rangle_{2, \omega} \\ &= 4 \int_{\mathbb{R}^n} \omega^{-1} A(t, \cdot) \nabla_x U_p(t) \cdot \nabla_x (U_p(t) - |U_p(t)|) \, d\omega \geq 0. \end{aligned}$$

Integrating from s to t in this inequality, we see that $t \mapsto \|U_p(t) - |U_p(t)|\|_{2, \omega}^2$ is a non-increasing function. Since it vanishes at $t = s$, we have for all $t > s$, $U_p(t) = |U_p(t)|$, that is $\Gamma_p(t, s)f = |\Gamma_p(t, s)f|$, hence $\Gamma_p(t, s)$ is a nonnegative operator.

Step 2: passing to the limit: using uniqueness in $L^2((s, \mathfrak{T}); H_{\omega}^1(\mathbb{R}^n))$ for any $\mathfrak{T} > s$ combined with the boundedness of $(U_p)_{p \geq 1}$ in $L^2((s, \mathfrak{T}); H_{\omega}^1(\mathbb{R}^n))$ provided by the energy equality, it is easy to check that, up to extracting a sub-sequence, $(U_p)_{p \geq 1}$ converges weakly to U when $p \rightarrow \infty$ in $L^2((s, \mathfrak{T}); L_{\omega}^2(\mathbb{R}^n))$ for any $\mathfrak{T} > s$, and therefore $U(t)$ is nonnegative for all $t \geq s$. $\square$

Combining Caccioppoli inequality in Lemma 4.4, a weighted Sobolev inequality [HKM18, Theorem 15.26] and the Moser's iteration principle, we have the following L^{∞} -estimate on nonnegative local weak solutions. For a proof, one can follow the classical scheme or see [Ish99, Proposition 2.1] with lower order coefficients equal to zero.

Lemma 6.2. *Let $(t_0, x_0) \in \mathbb{R}^{1+n}$ and $R > 0$. If u is a nonnegative local weak solution of $\mathcal{H}u = 0$ in a neighborhood of $Q_{2R}(t_0, x_0)$, then*

$$\operatorname{ess\,sup}_{Q_R(t_0, x_0)} u = \|u\|_{L^{\infty}(Q_R(t_0, x_0))} \leq B \left(\frac{1}{\mu(Q_{2R}(t_0, x_0))} \int_{Q_{2R}(t_0, x_0)} u^2 \, d\mu \right)^{1/2}$$

where $B = B(n, D, M, \nu) > 0$ is a constant. The same estimate holds for nonnegative local weak solution of $\mathcal{H}^*v = 0$.

By combining Lemma 6.2 above, Lemma 6.1 and Proposition 5.12, we obtain the following result.

Proposition 6.3. *The operator $\mathcal{H}$ admits a nonnegative generalized fundamental solution $\Gamma(t, x; s, y)$ with, for all $t > s$, almost everywhere pointwise Gaussian upper bound, that is,*

$$(6.4) \quad 0 \leq \Gamma(t, x; s, y) \leq \frac{K_0}{\sqrt{\omega_{t-s}(x)} \sqrt{\omega_{t-s}(y)}} e^{-k_0 \frac{|x-y|^2}{t-s}},$$

for almost every $(x, y) \in \mathbb{R}^{2n}$, where $K_0 = K_0(n, D, M, \nu) > 0$ and $k_0 = k_0(M, \nu) > 0$ are constants.